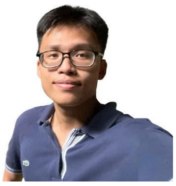

Tran Viet Thang

Robotics Engineering
VNU University of Engineering and Technology

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Profile Summary

• Research Assistant at VNU-UET with hands-on experience in humanoid and multi-robot systems. Specialized in researching and deploying deep learning models and large language models (LLMs) to optimize human-robot interaction/collaboration and autonomous task planning. Seeking an R&D position focused on task planning and learning for humanoid robots.

Education

•**VNU University of Engineering and Technology** *2023 – present*
B.Eng. in Robotics Engineering GPA: 3.2/4.0

Experience

•**VNU University of Engineering and Technology (VNU-UET)** *2024 – present*
Research Assistant, Department of Robotics Engineering Hanoi

- Research and develop AI algorithms to enhance human-robot interaction and human-robot collaboration (HRC) in humanoid systems.
- Explore LLM applications for task planning and real-time optimization in multi-robot environments.

•**AIBB Company** *11/2025 – present*
Robotics Engineer Hanoi

- Control and deploy motion policies, AI algorithms, and path-planning systems on the Unitree G1 humanoid robot.

Projects

•**Dual-arm Robot**
Developed an AI-integrated system for a dual-arm robot.

- Recognized and responded to human actions using LSTM models and path-planning algorithms.
- Executed human voice commands by leveraging LLMs to generate hierarchical task plans, incorporating perception and inverse kinematics for object manipulation.

•**Task Planning for Multi-Robot Collaboration**
Framework for multi-robot collaboration using LLMs, Directed Acyclic Graphs (DAGs), and online scheduling.

- Developed an LLM-based framework using Directed Acyclic Graphs (DAGs) to decompose high-level commands into hierarchical plans; implemented dynamic task allocation to minimize idle time and maximize parallel execution.

•**Unitree G1 Control and Policy Training**
Real-world deployment and experimentation on the Unitree G1 humanoid robot.

- Engineered advanced control systems for the Unitree G1 to execute high-precision manipulation and stable bipedal locomotion tasks.
- Integrated global and local path-planning algorithms to enable autonomous movement in dynamic environments.

– Trained action policies using reinforcement learning in simulation, achieving successful sim-to-real deployment on physical hardware.

• **Enhancing Human-Robot Interaction via Body Language**

Framework integrating LLMs, Dynamic Movement Primitives (DMP), and motion synchronization.

– Developed a synchronization framework to seamlessly align the robot's audio responses with natural body movements.
– Utilized DMP to generalize expressive actions from a motion library, and applied SMPL models to retarget human motion onto the humanoid robot.

Technical Skills

Languages: Python, C++, C, MATLAB
Robotics Middleware & Simulation: PyBullet, ROS2, MoveIt, Gazebo, MuJoCo, Isaac Sim/Lab
AI & Frameworks: PyTorch, Keras, OpenCV, NumPy
Developer Tools: Git/GitHub, Linux, Google Colab, PyCharm
Soft Skills: Teamwork, Research, English

Publications

[1] **Thang Tran Viet**, Thanh Nguyen Canh, Huy Uong Gia, Phuc Dinh Van, Son Tran Duc, Minh Ngoc Do, and Hoang Van Xiem, *"Development of a Humanoid Robot Prototype for Multimodal Human-Robot Interaction"*, RIVF International Conference on Computing and Communication Technologies (RIVF 2025), Ho Chi Minh City, Vietnam, Dec. 2025

[2] Thanh Nguyen Canh, **Thang Tran Viet**, Son Tran Duc, Trang Huyen Dao, Viet-Ha Nguyen, Xiem Hoang Van, *"Efficient Human-Robot Interaction via Deep Perception and Flexible Motion Planning"*, International Conference on Interactive Collaborative Robotics (ICR 2025), Hanoi, Vietnam, Nov. 2025

[3] **Thang Tran Viet**, Thanh Nguyen Canh, Phuc Dinh Van, Xiem Hoang Van, Nak Young Chong, *"OSDAG: Online Scheduling for Efficient Multi-Robot Collaboration"*, International Conference on Control, Automation and Systems (ICCAS 2026), 2026.

[4] **Thang Tran Viet**, Thanh Nguyen Canh, Huy Gia Uong, Phuc Dinh Van, Xiem Hoang Van, Nak Young Chong, *"WaveSync: Constrained Wavefront Optimization for Synchronized Co-Speech Gestures in Humanoid Robots"*, International Conference on Control, Automation and Systems (ICCAS 2026), 2026.